

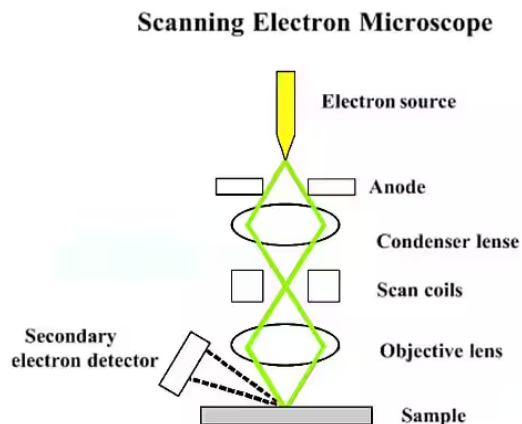
Scanning electron microscope (SEM)

How do SEMs work?

In an SEM, a focused beam of electrons is shot at a sample. The electrons interact with the sample, producing electrons, photons, and radiation as output. The following methods each measure a different type of output:

Backscattered Electron (BSE) Imaging

- Detects electrons, which collide elastically with atoms
- Large atoms are stronger scatterers of electrons, and thus appear lighter in the output than lighter atoms
- Useful for **distinguishing different elements** in an image

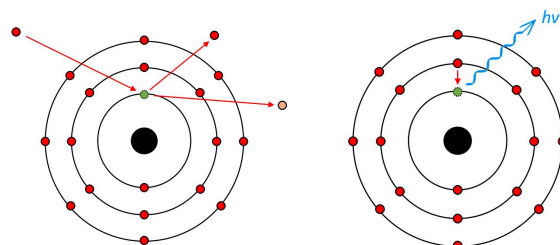


Secondary Electron (SE) Imaging

- Secondary electrons occur due to inelastic interactions with the atoms of the sample, and have lower energy than backscattered electrons
- SEs originate from the surface regions of the sample, making SE imaging good for **determining the topology** of a sample

Energy Dispersive Spectroscopy (EDS)

- EDS measures the X-rays emitted by a sample, when the primary electron beam displaces an inner core electron (see right).
- The energy of each X-ray is unique to each element. Thus, EDS is used for **elemental mapping** of a sample



The Breakerspace has two scanning electron microscopes (SEM), a ThermoFisher **Phenom XL** and **Phenom Pure**. What do you use each machine for?

Phenom Pure	Phenom XL
• For single samples, which can fit on a stud • For samples which need to be frozen (i.e. biological material)	• To image larger samples or multiple samples at once • To use EDS • For more vacuum and voltage levels
Both are outfitted with a backscatter electron detector (BSD) and secondary electron detector (SED).	

Instructions on using the Breakerspace SEMs

Sample Preparation

- Wear nitrile gloves to keep the SEMs clean.
- Prepare everything on the sample prep table.
- Use the conductive dot stickers to attach your sample to a metal stud.
- Flush your sample with compressed air to remove loose particles.
- For special samples:
 - Wet samples: dehydrate or use the cold stage. *Do NOT load wet samples into the machine!*
 - Non-conductive samples: use a low vacuum or gold-sputter
 - Samples which need polishing: polish by hand and/or use the ion mill



Instrument startup

- Log on to the computer using your MIT Kerberos
- Start the Phenom User Interface software
- If the instrument is not connected, connect in Settings>Status

Instrument operation

- While wearing gloves, load your stud onto the sample holder, and load the sample holder into the SEM
- After closing the door, the Phenom User Interface will open onto an optical view of the sample.
- To move to the SEM view, press the button with the magnifying glass on the top left.
- Perform characterization relevant to your sample (image capture, measurement/annotation, EDS in Phenom XL)
- Unload your sample while wearing gloves.

Instrument shut-down

- Close the software - press F11 to switch out of fullscreen view
- Log off Windows

See the Breakerspace website for the extended SOP! Each workstation also has a desktop icon linking to its SOP.

As always, if you need help, contact a lab assistant!